

Subject Description Form

Subject Code	ITC526
Subject Title	Computing Systems for Garment Manufacture
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	This subject aims at providing students with a solid comprehensive knowledge of the theories and application of the state-of-the-art computing system used in the fashion and clothing industry. On completion of this subject, the students will be able to: 1. implement computing systems in their related work; 2. evaluate the capabilities and limitations of the application for each type of computing systems; 3. understand the related management issues, such as training and work flow.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: 1. Demonstrate a deep understanding of the state-of-the-art computer related technologies in the international fashion and textile supply and distribution activities. 2. Exercise analytic skills and propose solutions in complex problems at managerial levels. - Select the right computer system to solve their problem. - Capable to analyze the existing systems available in the market. - Capable to adopt the computer system with proper training. - Capable to prepare a procurement plan, both financially and technically. 3. Demonstrate their potential and enthusiasm as active learners and motivate themselves for continuing development personally and professionally - Capable to plan for starting a new business that is based on a new computerized technology.

Subject Synopsis/ Indicative Syllabus	1. <u>Computer Aided Geometric Design</u> Curve model, surface model, continuity 2. <u>Computer Aided Fashion Design</u> 2D illustration, texture mapping, draping, virtual fashion 3. <u>Computer Aided Body Measurement Chart</u> 3D image scanning, generation of 3D virtual mannequin, 3D automated body measurement, automated size chart builder 4. <u>Computer Aided Pattern Design</u> 2D pattern, 3D pattern, grading, marker, optimized nesting 5. <u>Computer Aided Pattern Cutting</u> Cutting path 6. <u>Manufacturing Knowledge Portal, Merchandising Software, Costing Software, Evaluation and management of CAD/CAM systems</u> 7. <u>Communication System</u> Web Conferencing system, Teleconferencing system, EPAS prototyping visualization system 8. <u>RFID</u> Active and passive type, application in warehouse and retail shop 9. <u>Artificial Intelligent Systems</u> Applications of ANN, FL and GA in clothing industry 10. <u>Business Venture Planning</u> Writing funding proposal and investment proposal.
Teaching/Learning Methodology	It provides a comprehensive understanding of the computing systems used in the fashion and clothing industry. Theory and practice are both included in this subject. To enhance the understanding of up-to-date computing systems are actually implemented, the state-of-the-art computing systems used in fashion and clothing industry will be introduced. Students are expected to understand the theories of the state-of-the-art computing systems and how to implement in the fashion and clothing industry through the assignments and projects.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weightin g	Intended subject learning outcomes to be assessed				
			1	2	3	4	5
	1. Coursework	50%	✓	✓	✓	✓	✓
	2. Examination	50%	✓	✓	✓	✓	✓
	Total	100%					
	Coursework includes both a group presentation and a written assignment, which can be combined to be a comprehensive procurement plan. The examination is needed to test the general understanding of the functionality, limitation and potential of the different computer systems, the ability to evaluate a procurement plan and implement a business plan.						
Student Study Effort Expected	Class contact:						
	▪ Lecture					28 Hrs.	
	▪ Tutorial					11 Hrs.	
	▪ Self-study					66 Hrs.	
	Total student study effort					105 Hrs.	
Reading List and References	<u>Indicative Reading List and References:</u>						
	1. Efrat, S., The Development of a Method for Generating Patterns for Garment That Conform to the Shape of the Human Body, Ph.D. Thesis, Leicester Polytechnic, 1982.						
	2. Farin G., Curves and Surfaces for Computer Aided Geometric Design, A Practical Guide, 3 rd Edition, Academic Press, San Diego, 1993.						
	3. Hastings, Ashley J., CAD at work : making the most from computer-aided design, McGraw-Hill, New York, 1997.						
	4. Miller, P. B., Autocad For The Apparel Industry, Delmar Publishers, 1994.						
	5. Ng R., Computer Modelling for Garment Pattern Design, PhD Thesis, PolyU, 1998.						
	6. Price J. and Zamkoff B., Grading Techniques For Modern Design, 6 th Edition, Fairchild Publication, 1990.						

	7. Steinhaus, N.H. and Lott I.M., I Cad--can You? : Autocad For Apparel Design And Pattern Development, Grand Rapids Publications, 1991. 8. Trouvat P., Grading Manual, ESMOD Edition, 1995. 9. Uchiyama M.N.C. and Trouvat P., Pattern Making Manual Womens Garments, ESMOD Edition, 1997. 10. Yu, W.M., Harlock, S.C., Yeung, K.W., "Contour Measurements of Moulded Brassiere Cups Using a Shadow Moiré Technique", Proceedings of The 3rd Asian Textile Conference, Vol. 1, page 300-308, 1995. 11. User Menu, Prima System 12. OptiTex User Menu, Scanvec Garment Systems 13. User Menu, Process-based Accounting System (Prototype version), ITC, PolyU 14. Apparel Knowledge Portal Site (URL to be announced in class) 15. Lecture Materials on RFID, AI
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